NEATON ANG

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PROFILE

Data Scientist with 2+ years of experience turning ambiguous, high-stakes problems into testable hypotheses and deployable ML/AI solutions. Skilled at rapidly benchmarking classical ML, LLM, and embedding-based approaches to identify what actually moves the needle, and at communicating trade-offs across performance, interpretability, and cost to both technical and business stakeholders. Applied experience spans defense, healthcare, and generative AI, partnering cross-functionally to move work from research prototype to production.

EDUCATION

Columbia University, New York, NY **Sep 2025 – Dec 2026**

MS Data Science (GPA: 3.73/4.00)

Selected Applied ML Projects:

- **Hybrid Agreement Clustering** – Prof. J. Yang: Designed and evaluated high-dimensional clustering algorithms using BERTopic and LLM-based evaluators; benchmarked configurations via statistical validation and quality metrics (Grade: A)
- **Agentic Trading from Macroeconomic News** – Prof. A. Capponi: Built end-to-end agentic pipelines to extract sentiment-based factors from macro news; backtested market impact and predictive signals (Grade: A)
- **AI for Science: Clinical ML** – Dr. K. Krstovski: Research work: Benchmarking classical ML vs. fine-tuned Transformers for cancer detection in EHRs; grew into the funded VA-collaboration research role below

National University of Singapore (NUS), Singapore **Aug 2019 – May 2023**

B. Hons Data Science & Analytics (CGPA 4.25/5.00)

- NUS Sports Scholar (Badminton); Nominee, Lee Hsien Loong Outstanding All-Round Achievement Award
- Semester Exchange at Technical University of Munich; NUS Office of Admissions Campaign Ambassador

PROFESSIONAL EXPERIENCE

Columbia University – Data Science Institute, New York, NY **Jan 2026 – Present**

Graduate Research Assistant, Clinical NLP - in collaboration with the U.S. Department of Veterans Affairs (Advisor: Dr. K. Krstovski)

- Framed detection of metastatic prostate cancer (mPCa) in electronic health records as a binary classification hypothesis that unstructured clinical text would outperform structured lab data alone, benchmarking mPCa vs. localized disease across radiology notes and discharge summaries
- Ran systematic language analysis and comparative experiments across TF-IDF text representations, numeric lab features, and combined modalities using classical ML models; best approach (XGBoost, combined notes) achieved 0.89 sensitivity / 0.85 specificity
- Extending the pipeline with LLM and embedding-based text representations, testing alternate note sections and document structures to evaluate incremental gains over the TF-IDF baseline
- Co-authoring a paper on the automated mPCa detection framework for submission to a conference venue

DSO National Laboratories, Singapore **Dec 2023 – Sep 2025**

Member of Technical Staff – Machine Learning Researcher (Data Team)

- Part of the team pre-training multilingual LLMs (3B–70B parameters) from scratch; formed and tested hypotheses on which data-selection strategies meaningfully outperformed statistical baselines across languages and domains (Southeast Asia focus)
- Designed controlled experiments to quantify trade-offs between performance, robustness, and computational cost across candidate approaches, including fine-tuning small language models as context-aware data classifiers
- Worked with trillion-token-scale (multi-terabyte) unstructured datasets; re-architected preprocessing pipelines for a ~12× throughput improvement via parallelization
- Partnered cross-functionally to translate experimental findings into scalable, production-ready data pipelines

Ethlas / Eleos Labs, Singapore **May 2023 – Dec 2023**

Data Scientist – Generative AI

- Spearheaded the company's Gen AI initiatives as sole data scientist, advising leadership on which use cases were worth building given technical feasibility, risk, and ROI
- Led fine-tuning of a Stable Diffusion model for automated game-asset generation, cutting manual design workload by 65%
- Built LLM-powered NPC personas (RAG, LangChain) enabling open-ended, in-context player conversations
- Developed a fraud-detection model (88% recall) through feature engineering and iterative model validation

DSO National Laboratories, Singapore **Dec 2022 – May 2023**

Machine Learning Intern – Natural Language Processing

- Built a fact-checking / fake-news detection pipeline testing LLMs (GPT-3, BERT, BLOOM, FLAN-T5) across zero-shot, few-shot, and fine-tuned approaches, augmented with live web-retrieved evidence; prompt-engineering iterations lifted accuracy by 5%

LEADERSHIP & ATHLETICS

- **NUS Badminton, Team Captain** (2020–2022): Represented Singapore at the World University Games 2023; Singapore University Games Champion (2019, 2022); grew NUS Recreational Badminton to 5,500+ members
- **NUS Students' Sports Club Management Committee**: Vice-Project Director of Marketing for the Inter-Faculty Games, the largest institution-wide student event

TECHNICAL & ANALYTICAL SKILLS

Analytics & Modeling: Machine Learning, Predictive Modeling, Hypothesis Testing & Experimental Design, NLP, LLMs & Embedding Models, High-Dimensional Modeling

Programming & Frameworks: Python, R, SQL, Java, PyTorch, TensorFlow, Hugging Face, LangChain, scikit-learn, XGBoost

Data & Platforms: AWS, Distributed Computing (Slurm), Big Data Processing